

An Exact Algorithmic Solution of the Belgian Chocolate Problem

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Abstract

For $\delta > 0$, the Belgian Chocolate Problem asks for nonzero real Hurwitz stable polynomials x, y, p with $\deg x \geq \deg y$ and $p = (s^2 - 2\delta s + 1)x + (s^2 - 1)y$. We specify an exact algorithm that computes a critical real Q by strict rational enclosures of any prescribed precision and prove that the original admissible set is $(0, (1 - Q)/(1 + Q))$. The analytic parameter set includes its endpoint Q , but the strict polynomial parameter set does not. The proof combines explicit uniform bounds for four analytic factors, a finite semialgebraic hierarchy with an all-orders realization theorem, finite certificates on both strict sides, and a two-test-point terminating subdivision algorithm. Finite polynomial realization uses strict parameter slack; endpoint exclusion uses an explicit polynomial perturbation that preserves actual degrees even in the presence of leading-term cancellation. No degree bound, closed-form value, or practical complexity estimate is assumed.

1 Introduction

The problem considered here is the original polynomial simultaneous-stabilization problem stated in Section 2. Its quantifiers range over all finite degrees, and stability is strict. We determine its entire admissible parameter set by specifying a real number through a total certified approximation algorithm and proving the exact interval criterion, including the excluded endpoint.

The proof distinguishes two issues. Open-disc analytic feasibility is closed at its lower threshold; strictly stable polynomial feasibility is open there. The analytic hierarchy therefore cannot be identified pointwise with original polynomial feasibility at the threshold. The precise interface is proved in Sections 9–10.

The algorithm uses only integer and rational arithmetic, finite polynomial enumeration, and effective real-algebraic decisions. Its output defines the threshold independently of a feasibility query for the original problem. Theorem 11.1 is the final combined statement.

2 Original simultaneous-stabilization problem

Definition 2.1. A nonzero $h \in \mathbb{R}[s]$ is *Hurwitz stable* if every complex zero λ satisfies $\operatorname{Re} \lambda < 0$. Nonzero constants are stable. Put $a_\delta(s) = s^2 - 2\delta s + 1$ and $b(s) = s^2 - 1$. Define

$$\mathcal{A} = \left\{ \delta > 0 : \begin{array}{l} x, y, p \in \mathbb{R}[s] \setminus \{0\} \text{ are stable, } \deg x \geq \deg y, \\ p = a_\delta x + by \end{array} \right\}. \quad (1)$$

The symbol p denotes the third polynomial; z will denote a disc variable.

There is no leading-coefficient sign or monic normalization. The actual degree of p can be smaller than $\deg x + 2$.

Lemma 2.2. *If $\delta \in \mathcal{A}$, then $0 < \delta < 1$.*

Proof. At $\delta = 1$, evaluating the identity at $s = 1$ gives $p(1) = 0$. For $\delta > 1$, the roots $r_{\pm} = \delta \pm \sqrt{\delta^2 - 1}$ of a_{δ} satisfy $0 < r_- < 1 < r_+$. Evaluation gives $p(r_{\pm})/y(r_{\pm}) = r_{\pm}^2 - 1$, with opposite signs. But p/y is a continuous nonzero real function on $(0, \infty)$, so it has constant sign. Both cases contradict stability. $\square$

Lemma 2.3 (Parameter change). *On $I = (0, 1)$ the maps*

$$q(\delta) = \frac{1 - \delta}{1 + \delta}, \quad \delta(q) = \frac{1 - q}{1 + q} \quad (2)$$

are mutually inverse strictly decreasing bijections.

Proof. Substitution gives the inverse identities. Each map takes I to I and has derivative $-2/(1+x)^2 < 0$. $\square$

Write $P(q)$ for $\delta(q) \in \mathcal{A}$, for $q \in I$.

3 Analytic reformulation

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. For $0 < t < 1$, let $\mathcal{F}_t$ consist of holomorphic $f : \mathbb{D} \rightarrow \mathbb{C}$ such that $f(\bar{z}) = \overline{f(z)}$, the only zero is the simple zero 0, and the only 1-points are the two simple points $it, -it$. Put

$$A(q) : \mathcal{F}_{\sqrt{q}} \neq \emptyset, \quad \mathcal{T} = \{q \in I : A(q)\}, \quad q_* = \inf \mathcal{T}.$$

Nonemptiness and a positive lower bound for $\mathcal{T}$ are proved below. The symbol q_* is used in proofs, not in the algorithm's instructions.

Lemma 3.1 (Polynomial-to-analytic map). *For every $q \in I$, $P(q)$ implies $A(q)$.*

Proof. For a polynomial witness put

$$w = \frac{s-1}{s+1}, \quad s = \frac{1+w}{1-w}, \quad f(w) = \frac{b(s)y(s)}{p(s)}.$$

The second map is a biholomorphism from $\mathbb{D}$ onto the right half-plane. Direct algebra gives

$$(s+1)^2 w = b(s), \quad (s+1)^2 (w^2 + q) = (1+q)a_{\delta(q)}(s). \quad (3)$$

Stability makes f holomorphic, real symmetric, and gives $1 - f = a_{\delta(q)}x/p$. The only right-half-plane zero of b is the simple zero 1, corresponding to $w = 0$. The two distinct roots $\delta \pm i\sqrt{1 - \delta^2}$ of a_{δ} are in that half-plane and correspond to $w = \pm i\sqrt{q}$. Neither x, y nor p has a zero there. The change of variable has nonzero derivative, so the specified fibres remain simple and there are no additional fibres. The boundary point $w = 1$ is not used; no assertion about $\deg p$ is needed. $\square$

Lemma 3.2 (Four-factor representation). *For $q \in I$, $A(q)$ is equivalent to the existence of four real-symmetric holomorphic functions u, v, U, V on $\mathbb{D}$ satisfying*

$$zu + (z^2 + q)v = 1, \quad uU = 1, \quad vV = 1. \quad (4)$$

Proof. From f take $u = f/z$, $v = (1 - f)/(z^2 + q)$, $U = 1/u$, $V = 1/v$. The quotients have removable singularities at the specified points; the exact simple fibres make them nonvanishing everywhere on $\mathbb{D}$. Conversely let $f = zu$. The inverse identities make u, v nonvanishing. Thus f has just the simple zero 0, and $1 - f = (z^2 + q)v$ has exactly the two specified simple zeros. Symmetry is preserved in both directions, equivalently all Taylor coefficients are real. $\square$

4 Uniform bounds and effective coefficient boxes

4.1 Classical analytic inputs

Poincaré densities are normalized to curvature -1 , so $\rho_{\mathbb{D}}(z) = 2/(1 - |z|^2)$. We use the following standard theorems in the indicated forms.

Schwarz–Pick. If $h : G \rightarrow H$ is holomorphic between hyperbolic plane domains, then $\rho_H(h(z))|h'(z)| \leq \rho_G(z)$.

Schottky–Hempel density estimate. For $Y = \mathbb{C} \setminus \{0, 1\}$,

$$\rho_Y(w) \geq \frac{1}{2|w|(C + |\log |w||)}, \quad C = \frac{\Gamma(1/4)^4}{4\pi^2}.$$

This is the weakened form of the usual estimate in this normalization. The Gamma integral gives $\Gamma(1/4) < 4 + e^{-1} < 5$. Since $\pi > 3$, $C < 625/36 < 18 < 64$, and hence

$$\rho_Y(w) \geq \frac{1}{2|w|(64 + |\log |w||)}. \quad (5)$$

Maximum modulus, argument principle, and Cauchy estimate. For a function holomorphic near a closed disc, its maximum modulus is bounded by its boundary maximum. If it has no boundary zero, the image boundary's winding number about zero equals the number of interior zeros, with multiplicity. A boundary bound M at radius r gives Taylor coefficient bound Mr^{-j} .

Gronwall. If $a \geq 0$ is absolutely continuous on $[0, \ell]$ and $a' \leq ba$ almost everywhere, with b integrable, then $a(s) \leq a(0) \exp(\int_0^s b(\tau) d\tau)$.

The density estimate and Schwarz–Pick apply to arbitrary holomorphic maps omitting $0, 1$; no covering or local-univalence condition is imposed.

4.2 An explicit all-functions bound

Fix $\beta = 3/4$. For rational $0 \leq r < 1$, define

$$\begin{aligned} R &= \frac{1 + \max\{r, \beta\}}{2}, & d &= \frac{1}{2} \min\{R - \beta, 1 - R\}, \\ J &= \lceil 16R/d \rceil + 1, & B &= 3^{66 \cdot 9^J}, \\ M(r) &= \left\lceil B \left(\frac{1}{R} + \frac{1}{R^2 - \beta^2} + R + R^2 + \beta^2 \right) \right\rceil. \end{aligned} \quad (6)$$

All denominators are positive and these are effective rational or integer data, independent of q_* and of all function degrees.

Lemma 4.1 (Uniform four-factor bound). *For $0 < t \leq \beta$, $f \in \mathcal{F}_t$, and its four factors,*

$$\max\{|u(z)|, |v(z)|, |U(z)|, |V(z)|\} \leq M(r) \quad (|z| \leq r). \quad (7)$$

Proof. On the circle $\gamma(\theta) = Re^{i\theta}$ the image $f \circ \gamma$ has winding numbers 1, 2 about 0, 1, respectively. It meets

$$K = \{w : 1/4 \leq |w| \leq 2, |w - 1| \geq 1/4\}.$$

Otherwise its connected image lies in one of the three disjoint regions $|w| < 1/4$, $|w - 1| < 1/4$, $|w| > 2$. The first two force one winding number to vanish. In the third the winding numbers about 0, 1 agree, since the segment $[0, 1]$ avoids the curve. Each is a contradiction. Thus some z_0 on the circle satisfies

$$1/4 \leq |f(z_0)| \leq 4, \quad 1/4 \leq |1 - f(z_0)| \leq 4. \quad (8)$$

For every center c on this circle, the disc of radius d is contained in $\mathbb{D} \setminus \{0, it, -it\}$: all deleted points and the unit boundary are at distance at least $2d$. There $h = f$ and $h = 1 - f$ both omit 0, 1. Schwarz–Pick and (5) give

$$\frac{|h'(z)|}{|h(z)|} \leq 2(64 + |\log |h(z)||) \frac{2d}{d^2 - |z - c|^2}.$$

Along a radial segment of length at most $d/2$, $|\log |h||$ is absolutely continuous. The derivative inequality and Gronwall imply

$$64 + |\log |h(z)|| \leq 9(64 + |\log |h(c)||), \quad |z - c| \leq d/2, \quad (9)$$

because $\exp(2 \int_0^{d/2} 2d/(d^2 - s^2) ds) = \exp(2 \log 3) = 9$. The circumference $2\pi R < 8R$ can be traversed from z_0 in at most J steps of chord length at most $d/2$. Applying (9) at their successive centers, using $\log 4 < 2$ and $e < 3$, yields

$$|f|, |1/f|, |1 - f|, |1/(1 - f)| \leq B \quad (|z| = R).$$

Since $|z^2 + t^2| \geq R^2 - \beta^2$ there, the respective factor bounds on this circle are

$$B/R, \quad B/(R^2 - \beta^2), \quad RB, \quad (R^2 + \beta^2)B. \quad (10)$$

All four factors are holomorphic on the entire disc. Apply the maximum modulus principle separately and use their sum in (6). $\square$

4.3 Rational coefficient boxes

For $m \geq 1$ and $j \geq 0$, put

$$r_m = 1 - 2^{-m}, \quad M_m = \max\{1, M(r_m)\}, \quad C_j = \min_{1 \leq m \leq j+1} M_m r_m^{-j}. \quad (11)$$

Lemma 4.2 (Coefficient bounds and growth). *Each of the four Taylor coefficients at index j has absolute value at most C_j , for every $0 < t \leq \beta$ and $f \in \mathcal{F}_t$. Moreover*

$$\limsup_{j \rightarrow \infty} C_j^{1/j} \leq 1. \quad (12)$$

Proof. Cauchy's estimate gives each bound $M_m r_m^{-j}$ and hence their finite minimum. For fixed m , $C_j \leq M_m r_m^{-j}$ when $j \geq m - 1$. The upper root growth is at most r_m^{-1} . Let $m \rightarrow \infty$. $\square$

In particular every coefficient sequence in these boxes, not just one already known to arise from a feasible function, defines a holomorphic power series on $\mathbb{D}$.

5 Finite semialgebraic hierarchy

Definition 5.1. For $q \in [0, 9/16]$ and $N \geq 0$, $H_N(q)$ is feasibility of the following system in $4(N+1)$ real variables $(u_j, v_j, U_j, V_j)_{0 \leq j \leq N}$:

$$|u_j|, |v_j|, |U_j|, |V_j| \leq C_j \quad (0 \leq j \leq N), \quad (13)$$

$$u_{k-1} + v_{k-2} + qv_k = \mathbf{1}_{k=0} \quad (0 \leq k \leq N), \quad (14)$$

$$\sum_{i=0}^k u_i U_{k-i} = \sum_{i=0}^k v_i V_{k-i} = \mathbf{1}_{k=0} \quad (0 \leq k \leq N). \quad (15)$$

Negative indices mean zero, and $\mathbf{1}_{k=0}$ is one if $k = 0$ and zero otherwise. Absolute-value bounds mean the two corresponding linear inequalities. No constraint is imposed on an infinite tail by this definition.

Proposition 5.2 (Finite exact decisions). *For rational q in the defined range, $H_N(q)$ is a finite rational semialgebraic feasibility question, decidable by a terminating exact procedure. Restricting a solution at level $N+1$ gives one at level N . Also $H_0(0)$ is infeasible.*

Proof. The data C_j are rational, and all displayed constraints are polynomial equalities or inequalities in finitely many variables. Effective quantifier elimination for real closed fields decides such formulas. Restriction retains precisely the lower-index equations. At $q = 0$ the zeroth linear equation is $0 = 1$. $\square$

We use the standard effective real-algebraic theorem in its algorithmic form: a fixed exact implementation, for example cylindrical algebraic decomposition with real-root isolation, terminates and correctly decides each finite rational first-order real-closed-field sentence. Merely a nonconstructive choice of a truth value is not the algorithm used here.

6 All-orders realization and the analytic threshold

Theorem 6.1 (All-orders equivalence). *For $0 < q \leq 9/16$,*

$$A(q) \iff \forall N \geq 0 \ H_N(q). \quad (16)$$

Proof. Necessity follows from Lemmas 3.2 and 4.2. For sufficiency take the compact product

$$\mathcal{K} = \prod_{j=0}^{\infty} [-C_j, C_j]^4.$$

Countable products of compact intervals are compact in the coordinate topology. Let $E_N(q)$ be the subset imposing only equations (14)–(15) through index N . These are closed cylinder sets, because each equation is continuous in finitely many coordinates, and $E_{N+1}(q) \subseteq E_N(q)$. A finite feasible solution, with all remaining coordinates set to zero, proves $E_N(q) \neq \emptyset$. This extension is used only for nonemptiness of that cylinder, not to satisfy higher equations.

The nested compact-set theorem gives a common infinite coefficient object in $\bigcap_N E_N(q)$. By (12) its four power series converge absolutely on $\mathbb{D}$, locally uniformly, and are holomorphic there. Cauchy products recover (4) coefficient by coefficient. In particular $uU = vV = 1$ gives exact nonvanishing, rather than a limiting non-strict inequality. Lemma 3.2 recovers the simple fibres and real symmetry, and proves $A(q)$. $\square$

No surjectivity of projections between feasible solution spaces is used. The finite solutions are not chosen greedily to be compatible.

Lemma 6.2 (Explicit strict seed). *There are real rational polynomial factors at $q = 1/2$ that are zero-free on $\mathbb{D}$. Consequently $\mathcal{T}$ is nonempty and $q_* < 1/2$.*

Proof. Take

$$u = -\frac{1000}{1089}(1 + \frac{4}{5}z)^2(1 + \frac{17}{20}z), \quad v = \frac{2}{1089}(1089 + 1000z + 272z^2).$$

Expanding $u = -(1000 + 2450z + 2000z^2 + 544z^3)/1089$ verifies $zu + (z^2 + 1/2)v = 1$. The roots of u are $-5/4, -20/17$. The discriminant for v is $-184832 < 0$, and its conjugate roots have modulus $\sqrt{1089/272} > 1$. Both polynomials remain nonvanishing on some disc $|z| < R$ with $R > 1$. For $f = zu$, $f(Rz)$ then belongs to $\mathcal{F}_{\sqrt{1/2}/R}$, establishing a feasible parameter below $1/2$. $\square$

Lemma 6.3 (Scaling with strict slack). *If $0 < r < q < 1$ and $A(r)$, then $A(q)$. In fact there are real-symmetric holomorphic nonvanishing factors u, v on $|z| < \sqrt{q/r} > 1$ satisfying $zu + (z^2 + q)v = 1$ there.*

Proof. Take $f_0 \in \mathcal{F}_{\sqrt{r}}$, $c = \sqrt{r/q}$, and $F(z) = f_0(cz)$. On $|z| < 1/c$ this has exactly the simple zero 0 and the simple 1-points $\pm i\sqrt{q}$. The quotients F/z and $(1 - F)/(z^2 + q)$ are holomorphic nonvanishing factors on that disc. $\square$

Theorem 6.4 (Attained analytic threshold).

$$0 < q_* < 1/2, \quad \mathcal{T} = [q_*, 1). \quad (17)$$

Proof. The strict seed gives nonemptiness and the upper bound. At a feasible $0 < q \leq 9/16$, the zeroth identity and box give $qv_0 = 1$, $|v_0| \leq C_0$, and hence $q \geq 1/C_0$. A sequence of feasible parameters tending to the infimum can be chosen eventually below $1/2$. Thus $q_* \geq 1/C_0 > 0$. Scaling proves upward closure, and the defining property of the infimum then puts every $q > q_*$ in $\mathcal{T}$.

Choose feasible $q_n \rightarrow q_*$ eventually in $(0, 1/2)$. Their four coefficient sequences lie in the common compact metrizable space $\mathcal{K}$. In $[0, 1/2] \times \mathcal{K}$ take a convergent subsequence. Its parameter is q_* ; every finite coefficient equation passes to the limit by continuity. The power-series construction in the proof of Theorem 6.1 gives four factors at q_* . Since $q_* > 0$, the two 1-points stay distinct. Lemma 3.2 gives $A(q_*)$. $\square$

7 Finite certificates on both strict sides

Corollary 7.1 (Finite obstruction). *For $0 < q \leq 9/16$, $q < q_*$ if and only if some $H_N(q)$ is infeasible. For rational q this is a finite, exact all-degrees obstruction.*

Proof. Negate (16) and use (17). For rational q , apply Proposition 5.2. By Lemma 3.1, analytic impossibility also excludes any original polynomial witness. $\square$

Definition 7.2 (Positive certificate). For rational $q > 0$, a positive certificate is $v \in \mathbb{Q}[z]$ with $v(0) = 1/q$ for which

$$u(z) = \frac{1 - (z^2 + q)v(z)}{z} \quad (18)$$

and v are zero-free on $\mathbb{D}$. The constant condition makes u a rational polynomial and the factor identity exact.

This is decidable by real algebra: for $h \in \mathbb{Q}[z]$, write $h(a+ib) = H_1(a, b) + iH_2(a, b)$. Closed-disc nonvanishing is

$$\neg \exists a, b \in \mathbb{R} : a^2 + b^2 \leq 1, H_1(a, b) = H_2(a, b) = 0.$$

It contains only finite rational polynomial conditions.

Lemma 7.3 (Exact polynomial approximation). *If $0 < r < q < 1$ and $A(r)$, there are real polynomials u_*, v_* , zero-free on $\overline{\mathbb{D}}$, with $zu_* + (z^2 + q)v_* = 1$. When q is rational they can be chosen rational.*

Proof. Use the larger-disc factors from Lemma 6.3. Their minimum moduli m_u, m_v on $\overline{\mathbb{D}}$ are positive. Choose $0 < \eta < \frac{1}{2} \min\{m_v, m_u/(1+q)\}$. Taylor truncation approximates v within $\eta/2$ on $\overline{\mathbb{D}}$. For rational q , approximate its finitely many nonconstant real coefficients by rationals with sum of errors less than $\eta/2$, retaining $v_*(0) = 1/q$ exactly. For real q , retain the real Taylor polynomial. In either case $\|v_* - v\|_{\overline{\mathbb{D}}} < \eta$. Define u_* by (18), not by independent approximation. Its numerator vanishes at zero. On $|z| = 1$, $|u_* - u| \leq (1+q)\|v_* - v\|_{\overline{\mathbb{D}}}$. The difference is holomorphic at zero, so the maximum modulus theorem gives this bound throughout the disc. Both resulting polynomials therefore have positive minimum modulus. The identity, symmetry, and rationality assertions are exact. $\square$

The margins here depend on the individual strict pair $r < q$; no uniform estimate as $q \downarrow q_*$ is used. The existence proof does not require an algorithm to recover f_0 or its minimum moduli.

Theorem 7.4 (Positive-certificate characterization). *For rational $0 < q \leq 1/2$, a positive certificate exists if and only if $q > q_*$.*

Proof. Given a certificate, its two polynomial factors are nonvanishing on some disc of radius $R > 1$. Rescaling $f = zu$ to $f(Rz)$ gives $A(q/R^2)$, and hence $q_* \leq q/R^2 < q$. Conversely for $q > q_*$ choose $r \in [q_*, q)$ with $A(r)$ and apply Lemma 7.3. The same existence argument works for every real $q \in (q_*, 1)$, requiring rational coefficients only when q is rational. $\square$

8 Certified computation of the critical real

Fix one deterministic exact real-closed-field decision implementation as in Proposition 5.2. For $s \geq 0$, let $\mathcal{E}_s(q)$ be the finite list of rational polynomials of degree at most s , constant term $1/q$, and other coefficients in

$$\{a/b : a \in \mathbb{Z}, |a| \leq s, 1 \leq b \leq \max\{1, s\}\}.$$

Order by actual degree, then increasing lexicographic order of rational coefficient tuples, removing duplicates. Every rational polynomial with that constant term occurs for some finite s .

Start at $[L_0, R_0] = [0, 1/2]$, which strictly brackets q_* by Theorem 6.4. Given a current bracket $[L, R]$, put $a = (2L + R)/3$ and $b = (L + 2R)/3$ and execute the following staged search.

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for s = 0, 1, 2, ...:
  1. Decide H_s(a); if infeasible, return [a, R].
  2. Decide H_s(b); if infeasible, return [b, R].
  3. Test E_s(a) in its fixed order;
  if a positive certificate is found, return [L, a].
  4. Test E_s(b) in its fixed order;
  if a positive certificate is found, return [L, b].

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This implements parallel search by fair finite stages. Each stage contains only finitely many terminating decisions.

Theorem 8.1 (Total subdivision). *For rational $0 \leq L < q_* < R \leq 1/2$, the search terminates. Its output is a nested strict bracket of q_* , with width at most $2(R - L)/3$.*

Proof. The tests a, b are distinct positive rationals below $1/2$, so at most one equals q_* . The other has either a finite negative certificate by Corollary 7.1, or a finite positive certificate by Theorem 7.4. Exhaustiveness of the lists and the finite staging ensure discovery at some finite stage. No earlier stage can run forever. Negative certificates only raise the lower endpoint to a point strictly below q_* . Positive certificates only lower the upper endpoint to a point strictly above q_* . The four possible widths are $2/3, 1/3, 1/3, 2/3$ times the old width, respectively. $\square$

Theorem 8.2 (Algorithmic real and strict precision). *Iteration gives a fully specified sequence of nested rational brackets*

$$L_j < q_* < R_j, \quad R_j - L_j \leq \frac{1}{2}(2/3)^j. \quad (19)$$

Each stage is computable in finite time. Define Q to be the unique point of $\bigcap_j [L_j, R_j]$. Then $Q = q_$. For each $m \geq 0$ a terminating algorithm returns rational $L_m^{\text{prec}}, R_m^{\text{prec}}$ such that*

$$L_m^{\text{prec}} < Q < R_m^{\text{prec}}, \quad R_m^{\text{prec}} - L_m^{\text{prec}} < 2^{-m}. \quad (20)$$

Proof. Induction using Theorem 8.1 proves (19). Nested closed intervals with widths tending to zero have a unique common point; since each contains q_* , it is q_* . For a precision input m , choose the least j with $\frac{1}{2}(2/3)^j < 2^{-m}$ by finite rational comparisons, and return $[L_j, R_j]$. The geometric decay guarantees termination of this index search, and every subdivision is terminating. $\square$

The stage instructions use no q_* , Q , or original-feasibility oracle. Their correctness proof identifies the independently defined Q with the analytic infimum. Fixing a different correct real-algebraic engine does not change that limiting real. Define

$$\Delta = \frac{1 - Q}{1 + Q}. \quad (21)$$

9 Endpoint separation

Lemma 9.1 (Fixed actual degree stability). *For each nonzero stable h of actual degree d , there exists $\tau_h > 0$ such that every real polynomial g of degree at most d whose coefficients differ from those of h by less than τ_h has degree exactly d and is stable.*

Proof. For $d = 0$, use $|h(0)|/2$. Otherwise take disjoint closed circular discs around the distinct roots, all in the open left half-plane. Let Γ be their boundaries and

$$\mu = \min_{\Gamma} |h| > 0, \quad B_h = \sum_{j=0}^d \max_{\Gamma} |s|^j, \quad \tau_h = \min\{|h_d|/2, \mu/(2B_h)\}.$$

The leading coefficient stays nonzero, and $|g - h| < |h|$ on each circle. Rouché's theorem, for holomorphic functions near each closed disc, preserves the number of enclosed zeros with multiplicity. These counts total d ; the fixed degree leaves no other roots. Repeated roots are allowed. $\square$

Theorem 9.2 (Strict original-parameter perturbation). *If $\delta \in \mathcal{A}$, some $\delta' \in \mathcal{A}$ satisfies $\delta < \delta' < 1$, even when top coefficients of $a_\delta x + b$ cancel.*

Proof. By Lemma 2.2, $0 < \delta < 1$. Fix a witness and put $n = \deg x$, $m = \deg y \leq n$. If $m < n$, then $\deg p = n + 2$ with leading coefficient that of x . Keep x, y fixed and set $\delta' = \delta + e$, $p' = p - 2esx$. The identity is exact and the degree remains $n + 2$. If $M_x > 0$ is the maximum absolute coefficient of x , any sufficiently small $0 < e < \min\{(1 - \delta)/2, \tau_p/(4M_x)\}$ works by Lemma 9.1.

If $m = n$, let $e > 0$ and define

$$k = \sqrt{1 + e^2 + 2e/\delta}, \quad \delta' = \frac{k\delta}{1 + \delta e}, \quad \phi(s) = ks - e. \quad (22)$$

Use mixing coefficients

$$\begin{aligned} \alpha_e &= 1 + e^2 + e(\delta + 1/\delta), & \beta_e &= e^2 + e/\delta, \\ \gamma_e &= e(1/\delta - \delta), & \theta_e &= 1 + e/\delta. \end{aligned} \quad (23)$$

Comparison of quadratic coefficients gives

$$a_\delta(\phi) = \alpha_e a_{\delta'} + \gamma_e b, \quad b(\phi) = \beta_e a_{\delta'} + \theta_e b. \quad (24)$$

All coefficient checks are explicitly

$$\begin{aligned} \alpha_e + \gamma_e &= k^2, & \alpha_e - \gamma_e &= 1 + e^2 + 2\delta e, & \alpha_e \delta' &= k(\delta + e), \\ \beta_e + \theta_e &= k^2, & \beta_e - \theta_e &= e^2 - 1, & \beta_e \delta' &= ke. \end{aligned}$$

Define

$$X = \alpha_e x(\phi) + \beta_e y(\phi), \quad Y = \gamma_e x(\phi) + \theta_e y(\phi), \quad P_1 = p(\phi). \quad (25)$$

Then $P_1 = a_{\delta'} X + bY$ exactly. As $e \downarrow 0$ the coefficients of X, Y, P_1 tend to those of x, y, p . For all e the first two degrees are at most n ; their leading coefficients tend to the original nonzero ones. The actual degree of $P_1 = p(ks - e)$ is exactly $\deg p$, since $k > 0$, regardless of how many leading terms have canceled originally. Apply Lemma 9.1 at degree bounds $n, n, \deg p$. For sufficiently small $e > 0$ all three polynomials are stable, with $\deg X = \deg Y = n$. One may take $e = 2^{-j}$ for any sufficiently large j so that all three finite coefficient deviations are below their positive tolerances. Thus the perturbation is genuinely nonzero.

Finally direct algebra gives

$$1 - (\delta')^2 = \frac{1 - \delta^2}{(1 + \delta e)^2}. \quad (26)$$

For $e > 0$ this lies strictly between 0 and $1 - \delta^2$, so $\delta < \delta' < 1$. Stability is asserted only for the small e chosen above. Real coefficients, actual degrees, and the original degree inequality are preserved without normalization. $\square$

Remark 9.3 (Why actual degree matters). At $\delta = 1/2$, $x = s + 3/2$, $y = -s - 1/2$, $p = s/2 + 2$ form a stable witness with nominal degree three and actual degree one. Changing only δ to $1/2 + e$ gives $-2es^2 + (1/2 - 3e)s + 2$, whose root product is $-1/e < 0$ for $e > 0$. It has a positive root. The composition in (25) preserves the actual degree instead; no root at infinity is disregarded.

Theorem 9.4 (Endpoint exclusion). *For $q \in I$, $P(q)$ implies $q > Q$. In particular $\neg P(Q)$.*

Proof. Perturb its original polynomial witness to $\delta' > \delta(q)$ by Theorem 9.2. The corresponding q' lies strictly between 0 and q . Lemma 3.1 gives $A(q')$, whence $Q = q_* \leq q' < q$. This is an actual polynomial perturbation before the infimum inequality is used, not an exclusion deduced from the definition of an infimum alone. $\square$

10 Return to original polynomials

Lemma 10.1 (Cayley polynomial realization). *Let $q \in I$ and let real polynomials u, v be zero-free on $\mathbb{D}$, with $wu + (w^2 + q)v = 1$. Put $n = \max\{\deg u, \deg v\}$. Then*

$$\begin{aligned} x(s) &= (1+q)(s+1)^n v \left(\frac{s-1}{s+1} \right), \\ y(s) &= (s+1)^n u \left(\frac{s-1}{s+1} \right), \quad p(s) = (s+1)^{n+2} \end{aligned} \quad (27)$$

are nonzero real stable polynomials with $\deg x = \deg y = n$ and $p = a_{\delta(q)}x + by$. Rational input data give rational coefficients.

Proof. For $h(w) = h_d \prod_{j=1}^d (w - \omega_j)$ with $d \leq n$,

$$(s+1)^n h \left(\frac{s-1}{s+1} \right) = h_d (s+1)^{n-d} \prod_{j=1}^d ((1-\omega_j)s - (1+\omega_j)). \quad (28)$$

Zero-freeness gives $|\omega_j| > 1$, so $\omega_j \neq 1$ and

$$\operatorname{Re} \frac{1+\omega_j}{1-\omega_j} = \frac{1-|\omega_j|^2}{|1-\omega_j|^2} < 0.$$

Any additional roots are -1 . Empty products include constants. The coefficient of s^n is $h(1) \neq 0$, proving the actual degree n . Apply this to u, v and use (3) to obtain

$$a_{\delta(q)}x + by = (s+1)^{n+2}((w^2 + q)v + wu) = (s+1)^{n+2}.$$

The equality, first computed for $s \neq -1$, holds as a polynomial identity. Denominator clearing uses only real, or rational, operations. $\square$

Theorem 10.2 (Off-endpoint realization). *If $0 < r < q < 1$ and $A(r)$, then $P(q)$. Consequently every real $q \in (Q, 1)$ is polynomial feasible, with a rational polynomial witness whenever q is rational.*

Proof. Apply Lemma 7.3 and then Lemma 10.1. For $q > Q$ take $r = Q$, using analytic attainment. No uniform estimate in $q - Q$ is needed. This applies on all of I , not just the range of the finite hierarchy. $\square$

11 Main theorem

Theorem 11.1 (Exact algorithmic solution). *The algorithm of Section 8 uniquely determines a real $Q \in (0, 1/2)$ and terminates for every requested strict precision (20). With $\Delta = (1 - Q)/(1 + Q)$, the original problem has exactly the admissible set*

$$\boxed{\mathcal{A} = (0, \Delta), \quad \Delta \notin \mathcal{A}, \quad 1/3 < \Delta < 1.} \quad (29)$$

For every $q \in (0, 1)$ the two predicates are precisely

$$\boxed{A(q) \iff q \geq Q, \quad P(q) \iff q > Q.} \quad (30)$$

Thus for $0 < q \leq 9/16$ the all-layer analytic equivalence remains valid including Q , but its polynomial counterpart is the strict-slack relation

$$P(q) \iff \exists r \in (0, q) \ A(r) \iff \exists r \in (0, q) \ \forall N \ H_N(r). \quad (31)$$

The first equivalence in (31) holds on all of I .

Proof. Theorem 8.2 proves the algorithmic assertions and identifies $Q = q_*$. Theorem 6.4 gives the first predicate in (30). Theorems 9.4 and 10.2 give the second. Lemma 2.2 excludes $\delta \geq 1$; the decreasing inverse change of parameter gives $q(\delta) > Q$ exactly when $0 < \delta < \Delta$. The bounds on Δ follow from $0 < Q < 1/2$. Strict slack follows either from the perturbation and forward map, or the explicit intervals; realization proves its reverse direction. For $q \leq 9/16$, every $r \in (0, q)$ is in the range of Theorem 6.1. $\square$

Corollary 11.2 (Relative closure and endpoint meaning). *In $I = (0, 1)$, the analytic feasible set $[Q, 1)$ is the relative closure of the polynomial feasible set $(Q, 1)$. At Q all $H_N(Q)$ are feasible, but no original polynomial witness exists. No endpoint factors can extend nonvanishingly to a disc of radius $R > 1$.*

Proof. The set and hierarchy assertions follow from the preceding theorems. If endpoint factors extended as stated, their identity would extend by the identity theorem, and $Rzu(Rz)$ would belong to $\mathcal{F}_{\sqrt{Q}/R}$, contradicting $Q = \inf \mathcal{T}$. Open-disc analytic attainment therefore does not supply a closed-disc polynomial certificate. The closure in $\mathbb{R}$ instead includes 1, corresponding to $\delta = 0$ outside the original domain. In $(0, \infty)$ the original set has relative closure $(0, \Delta]$. $\square$

12 Exactness, computability, and non-stabilization

12.1 No finite level stabilizes at the threshold

Put $S_N = \{q \in [0, 1/2] : H_N(q)\}$ and $\alpha_N = \min S_N$.

Theorem 12.1 (Strict finite-level non-stabilization). *Each S_N is nonempty compact semialgebraic and α_N is an effectively computable real algebraic number. Moreover*

$$0 < \alpha_N < Q, \quad \alpha_N \uparrow Q, \quad \forall N \exists M > N : \alpha_M > \alpha_N. \quad (32)$$

For every fixed N there is $\varepsilon_N > 0$ such that $H_N(q)$ holds whenever $|q - Q| < \varepsilon_N$. The neighborhood can be chosen inside the defined parameter range.

Proof. The joint constraints in q and the finite coefficient box are closed and bounded, so their projection is compact. The seed at $1/2$ makes S_N nonempty. Quantifier elimination and real-root isolation give its algebraic minimum effectively. The zeroth equation gives $\alpha_N \geq 1/C_0 > 0$. Restriction gives $S_{N+1} \subseteq S_N$. All-orders realization and the analytic interval give $\bigcap_N S_N = [Q, 1/2]$. If α is the increasing limit of the minima, then for each fixed N its tail lies in the closed set S_N , so $\alpha \in S_N$. Hence $\alpha = Q$.

For strictness, take endpoint factors u_*, v_*, U_*, V_* . The four positive individual bounds in (10) each lie strictly below their sum used in $M(r)$. Cauchy's estimate and the finite minimum in (11) therefore give strict inequalities for each true coefficient: $|u_{*,j}|, |v_{*,j}|, |U_{*,j}|, |V_{*,j}| < C_j$. For q near Q define analytic germs at zero by

$$v_q = v_* + 1/q - 1/Q, \quad u_q = \frac{1 - (z^2 + q)v_q}{z}.$$

The quotient is removable, $v_q(0) = 1/q$, and $u_q(0) = (q/Q)u_*(0) \neq 0$. Their inverses are holomorphic germs, whose every fixed finite collection of coefficients depends continuously on q . The three identities hold for the germs. At Q the finitely many box constraints have strict margin, so continuity keeps them inside their boxes on some neighborhood. Thus H_N has feasible points below Q , proving $\alpha_N < Q$. Convergence then forces a later strict increase for every fixed N . $\square$

The germs in this proof need not have zero-free inverses on the whole disc. No single ε or infinite box control is asserted. One can also search effectively for a later $H_M(\alpha_N)$ that is infeasible, using exact real-algebraic input representation and the all-orders obstruction. Non-stabilization does not affect the precision algorithm, and asserts nothing about other possible formulas for Q .

12.2 Exact determination and limits of membership algorithms

The unique interval-intersection definition of Q uses the complete specified program and its proved convergence, not a finite decimal approximation. Together with both directions of (29) and the independent endpoint exclusion, it determines the original feasible set exactly. Absence of a closed form, an algebraicity result, or a useful complexity bound does not weaken these statements. The computed brackets transform to

$$\frac{1 - R_j}{1 + R_j} < \Delta < \frac{1 - L_j}{1 + L_j}, \quad \frac{1 - L_j}{1 + L_j} - \frac{1 - R_j}{1 + R_j} = \frac{2(R_j - L_j)}{(1 + L_j)(1 + R_j)} \leq 2(R_j - L_j).$$

One extra bit of strict precision for Q suffices for any requested strict precision for Δ .

For an effectively given real input promised different from Δ , successively refining the input and threshold intervals eventually decides the strict comparison. For rational $\delta > 0$, $\delta \geq 1$ is impossible and $\delta \leq 1/3$ is feasible. In the remaining range the rational transform lies in $(0, 1/2)$ and the two certificate searches decide all nonendpoint inputs. The parameter explicitly specified as Δ is infeasible by the endpoint theorem. No irrationality of Q or Δ is claimed, and no total rational-input membership algorithm at a potentially rational endpoint is asserted.

This must be distinguished from the unconditional totality of the precision algorithm: every precision input terminates, with no nonendpoint promise. A general algorithm deciding membership from an arbitrary black-box real Cauchy name would face the discontinuity of the interval characteristic function at its endpoint; it is not a conclusion of this paper. The exact interval theorem and endpoint exclusion do not require such an equality detector.